PALEOMAGNETIC RESULTS FROM NORTHERN MAINE AND THE WESTERN LIMIT OF "AVALON" IN THE MID PALEOZOIC

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Abstract. Lower Devonian volcanic rocks from Chapman syncline near Presque Isle, Maine yield paleomagnetic directions unlike those determined for stable North America. Sites in four different igneous units of the Dockendorff Group give a pole position of 24° N latitude and 132° E longitude. These rocks are unmetamorphosed, and appear to be unaffected by the mid-Devonian Acadian orogeny. The pole agrees with other published data for upper Devonian rocks of eastern New England and the Canadian Maritimes. This indicates that the Avalon microcontinent may have extended further west than supposed on the basis of faunal and geologic data.

Introduction

Paleomagnetic studies on Devonian age rocks of eastern North America have recently become important in the reconstruction of the paleo-continental margin. Studies by Kent and Opdyke [1978], Irving [1979] and Van der Voo et al. [1979] have pointed out the noticeable discrepancy between pole positions from cratonic rocks of mid Paleozoic age and those from near the present continental margin. The suggestion was made by Kent and Opdyke [1978] that left-lateral motion along the eastern seaboard during Carboniferous time would account for the southerly position of these poles in the Devonian.

It has been suggested that eastern New England and parts of the Maritime provinces of Canada were not part of North America in the early Paleozoic [Wilson, 1966; Schenk, 1971; McKerrow and Ziegler, 1972; and others], although this division was done on geologic rather than paleomagnetic evidence. Called the 'Avalon Platform' or Avalon microcontinent; it has been given several different origins and tectonic histories [Schenk, 1971].

To better understand the tectonic behavior of this microplate, and to clarify its boundaries, a study was undertaken of early Devonian volcanics in northern Maine. These rocks were chosen for: 1) their position between the stable platform and the alleged microcontinent, 2) their age prior to the Acadian orogeny, but non-involvement in the orogeny, and 3) their present unaltered state.

Geologic Setting

Northern Maine was an area of deposition from middle Ordovician to early Devonian, with sporadic volcanics within the sedimentary sequence. All but the youngest strata are weakly metamorphosed, with abundant fossils retained, which allows for dating and interpretation of Paleozoic units not well understood in the rest of the Northern Appalachians [Boucot, 1968]. Early Devonian volcanics, lying disconformably on late Silurian sediments have been mapped in detail by Boucot et al. [1964] in the Presque Isle Quadrangle, northeastern Maine. These deposits, the Dockendorff Group, consist of the Hedgehog Formation, basalt flows and sediment lenses, overlain by the Edmunds Hill Andesite, Chapman Sandstone and Swanback Formation. Due to the high competence of this group [Pavlides, 1974] the Acadian orogeny was felt only in terms of folding and minor faulting. The lack of any medium or high grade metamorphism allows one to investigate magnetic directions obtained prior to the mid-Devonian orogeny.

Age of the Dockendorff Group has been well established. Studies on faunal collection from tuffaceous boulders in the Hedgehog Formation, and from fossiliferous zones in the Chapman Sandstone yield ages of New Scotland (upper Gedinnian) [Boucot et al., 1964]. Rb/Sr data on nine samples from the Hedgehog Formation provide an age of 406 ± myr [Fullagar and Bottino, 1970; corrected for new decay constant, P.D. Fullagar, personal communication, 1979].

Sampling and Laboratory Procedures

Six sites were sampled for paleomagnetic studies in the Presque Isle quadrangle (Fig. 1): one in the Hedgehog Formation (site 5), three in the Edmunds Hill Andesite (sites 3, 4a, 4b), one in a diabase related to the Hedgehog Formation (site 2), and one in an Ordovician (?) volcanic unit (site 1). At each site, nine to 19 samples were drilled and oriented in the field.

At least one sample per site was subjected to progressive alternating field demagnetization up to 80 mT. Behavior of a typical sample is shown in Fig. 2. Although most samples show marked drop in intensity and acquire stable directions with increasing demagnetization, a few samples showed sporadic behavior and did not reach stable endpoints. In an attempt to remove all secondary components, all samples were treated to low thermal demagnetization (200° C) before alternating field demagnetization at 20 and 40 mT. Samples which appeared unstable after final demagnetization, or had not lost a significant amount of their intensity were discarded from further calculations (7 samples).

Results

Previously reported data on the Dockendorff Group [Brown, 1978] did not adequately correct for the structural setting. Corrections needed for these sites include the unfolding of the Chapman syncline and removal of the plunge from site 5. The syncline has a sigmodial trace, with sites 1, 2, 3, 4a and 4b located in areas of zero plunge. Structural attitudes were determined from the geologic map of Boucot et al.,

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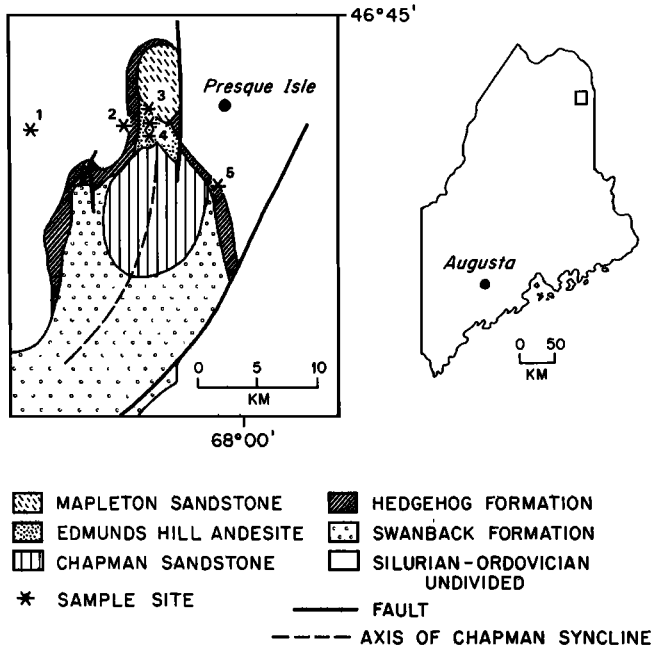


Fig. 1. Simplified geologic map of the Presque Isle area, Maine [after Pavlides, 1974].

[1964]. Uncorrected and corrected inclinations and declinations are shown in Fig. 3, with related statistics given in Table 1. The paleomagnetic results after unfolding are in good agreement and indicate that the magnetization was obtained prior to Acadian folding.

From a comparison of the final results it is obvious that site 1 is significantly different from the remaining sites, and from other recorded Devonian poles (Table 2). It is similar to published Ordovician poles from Quebec [Seguin, 1977], so the mapped age is considered correct. The results from site 3, even after demagnetization to 40 mT and tectonic corrections are widely scattered, and due to the exceedingly high alpha-95 value are not included in further calculations.

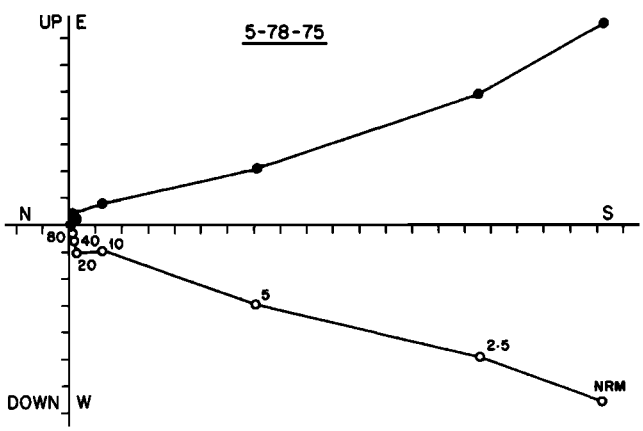


Fig. 2. Orthogonal demagnetization diagram [Zijderveld, 1967] of magnetization vector during demagnetization. Sample is from site 5 (Hedgehog Formation) before tectonic correction. Solid circles are projections on the horizontal plane, open circles are on the north-south vertical plane. Axes units are in amp/meter, demagnetization levels in mT.

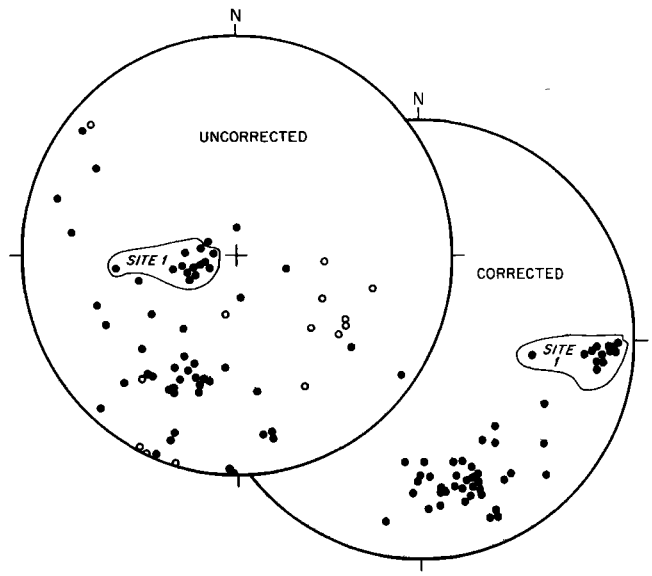


Fig. 3. Inclination and declination of demagnetized samples, before and after tectonic corrections. Site 3 is omitted from the plot after tectonic corrections.

Table 1 gives the mean poles calculated from acceptable sample and site directions, with a north magnetic pole at 24° N latitude and 132° E longitude.

Discussion

The suggestion that part of eastern New England and the Canadian Maritimes is exotic terrain has been put forth on faunal and geologic data [Wilson, 1966; Dewey, 1969; Schenk, 1971]. In these studies the boundary between the North American craton and the microplate has been associated with known fault systems close to the present continental coast (Fig. 4). Recently Kent and Opdyke [1978] have presented paleomagnetic data supporting this separation, and have suggested that the two plates did not join until late Carboniferous, and then along left-lateral strike slip faults. Support for this idea has come from Van der Voo et al., [1979] and (with modification) form Irving [1979]. Similar transcurrent motion has been proposed for the British Isles [Morris, 1976] which may or may not be on the same plate as Avalon [Schenk, 1971; Van der Voo, 1979].

Published data for Devonian rocks from the stable platform, the Avalon plate, and the intermediate area is presented in Table 2, with their present geographic locations shown in Fig. 4. It is obvious that the Dockendorff Group pole is aligned with the poles for Avalon, although poles from other locations in the intermediate area are not. It must be noted that all poles for Avalon are middle to upper Devonian in age.

Explanations for the agreement of the Presque Isle directions with those of the exotic terrain include: 1) the Chapman syncline has been altered in a way not taken into consideration in the tectonic corrections, and pole agreement is coincidental; 2) the Dockendorff Group represents another sliver or microplate, also in a southerly position in the Devonian, but not necessarily part

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TABLE 1. Magnetic directions before and after tectonic corrections

	N/S	H	I	D	I'	D'	K	A-95	Pole Position
Site 1 (NS, 90E)	13/14	20	68.9	259.9	17.3	94.1	69	5.0	3.7N, 12.7E
Site 2 (N10E, 90E)	8/9	20	27.8	232.2	32.4	151.7	11	17.1	20.9N, 140.7E
Site 3 (N10E, 50E)	15/15	40	-13.4	131.5	-28.7	169.8	2	35.4	57.4N, 266.8E
Site 4a (N10E, 50E)	14/14	20	42.8	202.5	33.8	159.1	96	4.1	22.2N, 133.2E
Site 4b (N10E, 50E)	5/5	20	34.7	204.1	31.1	168.5	102	7.6	25.7N, 124.1E
Site 5 (NS, 20S N155E, 90W)	11/16	20	12.0	185.7	29.0	164.2	20	10.5	26.3N, 129.0E
Mean Sites*	4				31.6	160.9	157	7.4	24.0N, 131.8E
Mean Samples*	38				31.7	160.5	25	4.6	23.8N, 132.2E

S is the number of samples collected, N is the number of samples used in site means
H is the demagnetization level in milliteslas
I and D are the magnetic inclination and declination before tectonic corrections
I' and D' are the magnetic inclination and declination after tectonic corrections
K and A-95 are the precision parameter and radius of cone of 95% confidence about the mean
Tectonic corrections (direction and angle of rotation) are given in parentheses for each site
*Mean sites and mean samples calculated without sites 1 and 3

TABLE 2. Devonian Paleomagnetic Poles, Eastern North America

Location	Age	Pole	Reference
North American Platform			
1 Columbus lm, Oh	Dm	45/120	Martin, 1975
2 Delaware lm, Oh	Dm	48/118	Martin, 1975
3 Catskill fm, Pa	Du	43/130	Phillips, 1966
4 Catskill fm, Pa	Du	44/124	Van der Voo et al., 1979
5 Catskill fm, Pa	Du	41/123	Liebes and Mc Donald, 1977
6 Catskill fm, NY	Du	47/117	Kent and Opdyke, 1978
Avalon Microcontinent			
9 Mafic complex, Ma	370	23/126	Schutts et al., 1976
11 Perry lavas, Me	Du	24/128	Phillips and Heroy, 1966
12 Perry lavas, NB	Du	26/109	Black, 1964
13 Perry seds, NB	Du	35/121	Black, 1964
14 Perry seds, NB	Du	32/118	Robertson et al. 1968
Intermediate Locations			
7 Belchertown Int, Ma	380	48/147	Ashwal and Hargraves, 1977
8 Hartwick Diorite, Ma	400	36/84	Brown and Kelly, 1977
10 Dockendorff Cp, Me	406	24/132	This study
11 Clam Bank Gp, Nfld	D1	28/146	Black, 1964
North American Platform Average, N=6		45/122	
Avalon Microcontinent Average, N=5		28/120	

of Avalon; 3) these rocks were formed on the same plate as the Avalon rocks, and have moved as suggested by Kent and Opdyke [1978]. The geologic information on the area [Boucot et al., 1964; Pavlides, 1974] is very adequate, and 1) is not considered likely. It is difficult at this time to distinguish between 2) and 3); detailed geologic information in northern and central Maine is scanty. Either possibility requires that the boundary between stable North America and the microplate(s) must be west of the Presque Isle quadrange. Further delineation will require addi-

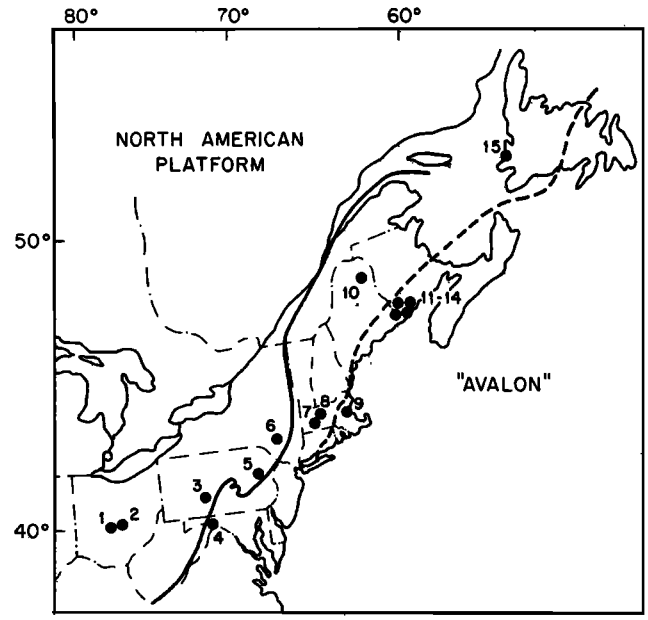


Fig. 4. Location of paleomagnetic studies on Devonian rocks of eastern North America. Numbers refer to data in Table 2. Solid line is western margin of the Appalachian orogen, dashed line is the western limit of Avalon [after Schenk, 1978].

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tional paleomagnetic studies in the Northern Appalachians.

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